MIL-N-23182A
26 June 1967
SUPERSEDING
MIL-N-23182(NAVY)
19 March 1962
(See 6.7)

MILITARY SPECIFICATION

NETWORKS, PULSE-FORMING, RADAR,

GENERAL SPECIFICATION FOR

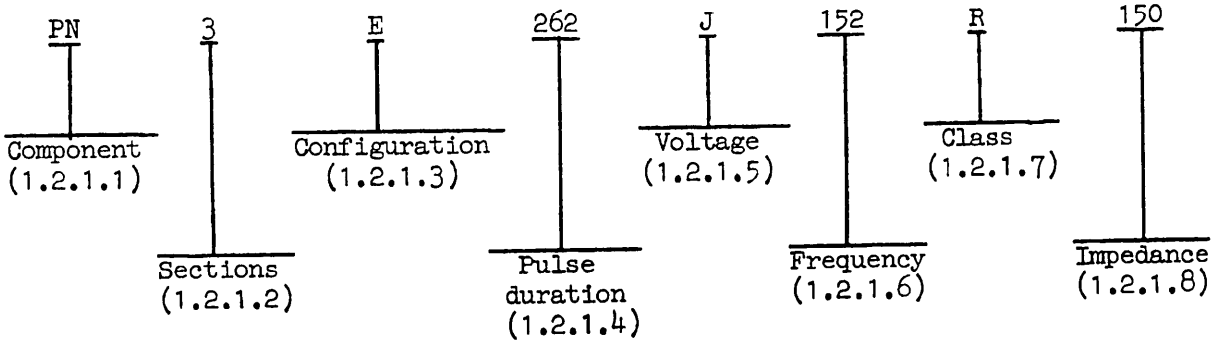
This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers pulse-forming networks primarily intended for use in military radar equipment. This specification is intended to cover both pulse-forming networks and pulse-forming networks combined with additional elements, such as despiking networks which are functionally related to the pulse-forming network. This specification is not intended to cover networks weighing more than 300 pounds or having direct current (dc) test voltage ratings greater than 50,000 volts (50 kilovolts (kv)), nor is it intended to cover open-frame or encapsulated units.

1.2 Classification.

1.2.1 Type designation. The type designation for pulse-forming networks shall be in the following form, and as specified (see 3.1 and 6.1):

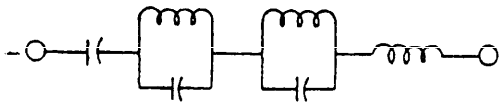


1.2.1.1 Component. Pulse-forming networks shall be identified by the two-letter symbol PN.

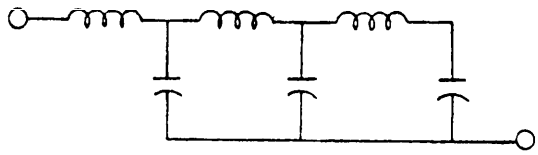
1.2.1.2 Sections. The sections shall be identified by a number which designates the applicable number of capacitance and inductance sections utilized by the network.

1.2.1.3 Configuration. The electrical type is the circuitry type of configuration as shown on figure 1, and designated by one of the following letters:

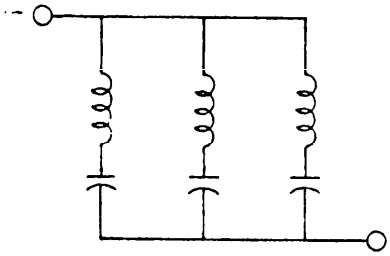
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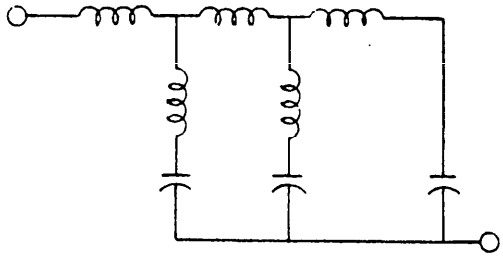
TYPE A



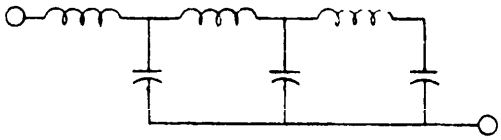
TYPE B
(SECTION CAPACITANCE NOT EQUAL VALUES)



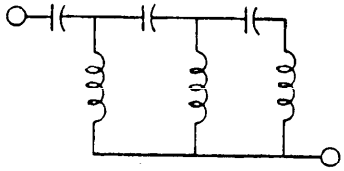
TYPE C



TYPE D



TYPE E
(EQUAL CAPACITANCE AND INDUCTANCE VALUES USED IN EACH SECTION)



TYPE F

ELECTRICAL CONFIGURATIONS OTHER THAN TYPES A THROUGH F

TYPE G 1/

1/ Electrical configuration for type 'G' shall be shown on individual specification sheets (see 3.1).

FIGURE 1. Electrical configurations of pulse-forming networks (three section types illustrated).

1.2.1.4 Pulse duration (see 6.6.1). The nominal pulse duration, expressed in nanoseconds, is identified by a three-digit number; the first and second digits designate significant figures and the third digit indicates the number of zeros to follow (tabulations in microseconds are given for information, in the specification sheets). For example: 262 indicates 2600 nanoseconds.

1.2.1.5 Voltage. The voltage shall be identified by a single letter in accordance with table I which designates the rated dc peak working voltage expressed in kv, to which the network is charged.

TABLE I. Voltage.

Symbol	kv	Symbol	kv	Symbol	kv	Symbol	kv
A	1	F	7.5	L	20	R	45
B	2	G	10.0	M	25	S	50
C	3	H	12.5	N	30		
D	4	J	15.0	P	35		
E	5	K	17.5	Q	40		

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1.2.1.6 Frequency. The frequency shall be identified by a three-digit number which designates maximum pulse repetition frequency (see 6.6.2) in pulses per second (pps). The first two digits represent significant figures and the last digit specifies the number of zeros to follow. For example: 152 indicates 1500 pps.

1.2.1.7 Class. The class of pulse-forming networks shall be identified by a single letter, in accordance with table II denoting the maximum operating temperature for the realization of the life expectancy specified (see 3.1). The maximum operating temperature is defined as the allowable ambient temperature plus the temperature rise developed within the network.

TABLE II. Class.

Symbol	Maximum operating temperature (°C)
Q	85
R	105
S	130
T	150
U	170
V	>170 as specified (see 3.1 and 6.1.2)

1.2.1.8 Impedance. The impedance shall be identified by a three-digit number which designates the nominal characteristic impedance (see 6.6.3) expressed in ohms. The first two digits represent significant figures and the last digit specifies the number of zeros to follow. For example: 150 indicates 15 ohms.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-571 - Solder; Tin Alloy; Lead-Tin Alloy; and Lead Alloy.
PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-601 - Box, Wood, Cleated-Plywood.
PPP-B-621 - Box, Wood, Nailed and Lock-Corner.
PPP-B-636 - Box, Fiberboard.
PPP-B-640 - Box, Fiberboard, Corrugated, Triple-Wall.

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